

Random Regressors and Moment-Based Estimation

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1 Linear Regression with Random x 's

The key of this chapter: Relax the assumption that the variable x is not random.

Modified Simple Regression Assumptions:

1. $y_i = \beta_1 + \beta_2 x_i + e_i$ correctly describes the relationship between y_i and x_i in the population, where β_1 and β_2 are unknown (fixed) parameters and e_i is an unobservable random error term.
2. The data pairs (x_i, y_i) , $i = 1, \dots, N$, are obtained by *random sampling*. That is, the data pairs are collected from the same population, by a process in which each pair is independent of every other pair. Such data are said to be independent and identically distributed (*iid*).
3. The expected value of the error term e , *conditional* on the value of x , is zero.
 - If $E(e|x) = 0$, then we can show that it is also true that x and e are uncorrelated, and that $Cov(x, e) = 0$. Explanatory variables that *are not* correlated with the error term are called *exogenous variables*.
 - Conversely, if x and e are correlated, then $Cov(x, e) \neq 0$ and we can show that $E(e|x) \neq 0$. Explanatory variables that *are* correlated with the error term are called *endogenous variables*.
4. In the sample, x must take at least two different values.
5. $Var(e|x) = \sigma^2$. The variance of the error term, conditional on any x , is a constant σ^2 .
6. The distribution of the error term is normal.
 - Assumption 2 states that both y and x are obtained by a sampling process, and thus are random. This is the only one new assumption on our list.

- The result that under the classical assumptions, and fixed x 's, the least squares estimator is the best linear unbiased estimator, is a finite sample, or a small sample. This means is that the result does not depend on the size of the sample.
- Under assumptions 1-6:
 1. The least squares estimator is unbiased.
 2. The least squares estimator is the best linear unbiased estimator of the regression parameters, and the usual estimator of σ^2 is unbiased.
 3. The distributions of the least squares estimators, conditional upon the x 's, are normal, and their variances are estimated in the usual way.

The usual interval estimation and hypothesis testing procedures are valid.

- If x is random, as long as the data are obtained by random sampling and the other usual assumptions hold, no changes in our regression methods are required.
- For the purposes of a "large sample" analysis of the least squares estimator, it is convenient to replace assumption 3 by:

3* $E(e) = 0$ and $Cov(x, e) = 0$.

- Now we can state that under assumptions 1, 2, 3*, 4, and 5, the least squares estimators:

1. Are consistent.

They converge in probability to the true parameter values as $N \rightarrow \infty$.

2. Have approximate normal distributions in large samples, whether the errors are normally distributed or not.

Our usual interval estimators and test statistics are valid, if the sample is large.

3. If assumption 3* is not true, and in particular if $Cov(x, e) \neq 0$ so that x and e are correlated, then the least squares estimators are inconsistent.

They do not converge to the true parameter values even in very large samples.

None of our usual hypothesis testing or interval estimation procedures are valid.

- The statistical consequences of correlation between x and e is that the least squares estimator is biased — and this bias will not disappear no matter how large the sample.
- Consequently the least squares estimator is inconsistent when there is correlation between x and e .

2 Cases in which x and e are Correlated

- When an explanatory variable and the error term are correlated, the explanatory variable is said to be *endogenous*.

This term comes from simultaneous equations models: It means "*determined within the system*".

- Using this terminology when an explanatory variable is correlated with the regression error, one is said to have an "*endogeneity problem*".
- The errors-in-variables problem occurs when an explanatory variable is measured with error.
- If we measure an explanatory variable with error, then it is correlated with the error term, and the least squares estimator is inconsistent.

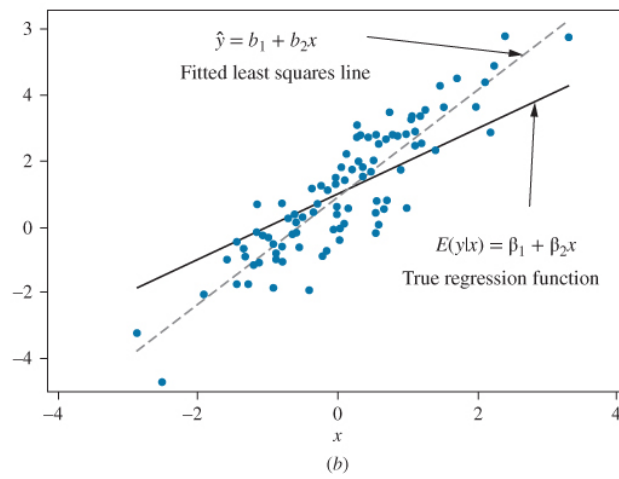
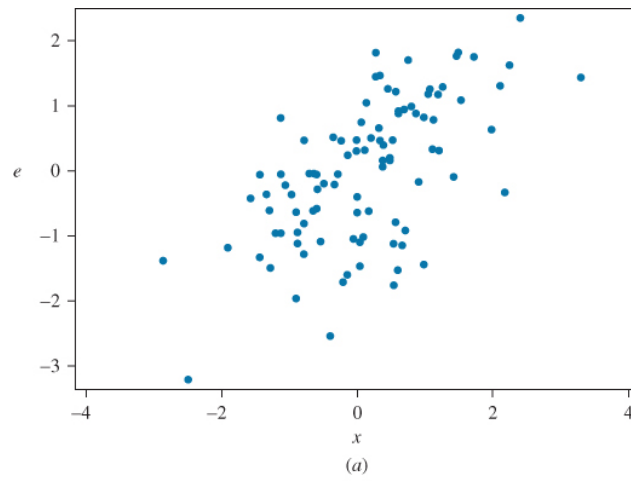


Figure 1: (a) Correlated x and e . (b) Plot of data, true and fitted regression functions

Example:

- Let y = annual savings, and let x^* = the permanent annual income of a person.
- A simple regression model is:

$$y_i = \beta_1 + \beta_2 x_i^* + v_i \quad (1)$$

- Current income (x_i) is a measure of permanent income, but it does not measure permanent income exactly. It is sometimes called a *proxy variable*.
- To capture this feature, let u_i = random disturbance, and define:

$$x_i = x_i^* + u_i \quad (2)$$

- Substituting x_i^* from 2 into 1:

$$\begin{aligned} y_i &= \beta_1 + \beta_2 x_i^* + v_i \\ &= \beta_1 + \beta_2 (x_i - u_i) + v_i \\ &= \beta_1 + \beta_2 x_i + (v_i - \beta_2 u_i) \\ &= \beta_1 + \beta_2 x_i + e_i \end{aligned} \quad (3)$$

- In order to estimate Eq. 3 by least squares, we must determine whether or not x is uncorrelated with the random disturbance e .
- The covariance between these two random variables, using the fact that $E(e) = 0$, is:

$$\begin{aligned} Cov(x, e) &= E(xe) = E[(x^* + u)(v - \beta_2 u)] \\ &= E[-\beta_2 u^2] \\ &= -\beta_2 \sigma_u^2 \\ &\neq 0 \end{aligned} \quad (4)$$

- The least squares estimator b_2 is an *inconsistent* estimator of β_2 because of the correlation between the explanatory variable and the error term.
- Consequently, b_2 does not converge to β_2 in large samples (even very large samples).
- In large or small samples b_2 is not approximately normal with mean β_2 and variance:

$$Var(b_2) = \frac{\sigma^2}{\sum_i (x_i - \bar{x})^2}$$

2.1 Simultaneous Regression Model

- Another situation in which an explanatory variable is correlated with the regression error term arises in simultaneous equations models (e.g., consider a supply and demand model where the equilibrium price and quantity are estimated simultaneously).
- Suppose we write:

$$Q = \beta_1 + \beta_2 P + e \quad (5)$$

- There is a feedback relationship between P and Q .
- Because of this, which results because price and quantity are jointly, or simultaneously, determined, we can show that $Cov(P, e) \neq 0$.
- The resulting bias (and inconsistency) is called the *simultaneous equations bias*.

- When an omitted variable is correlated with an included explanatory variable, then the regression error will be correlated with the explanatory variable, making it endogenous.
- Consider a log-linear regression model explaining observed hourly wage:

$$\ln(WAGE) = \beta_1 + \beta_2 EDUC + \beta_3 EXPER + \beta_4 EXPER^2 + e \quad (6)$$

- **Q:** What else affects wages? What have we omitted?
- **A:** Ability
- We might expect $Cov(EDUC, e) \neq 0$.
- If this is true, then we can expect that the least squares estimator of the returns to another year of education will be positively biased, $E(b_2) > \beta_2$, and inconsistent.
- The bias will not disappear even in very large samples.
- Estimating our wage equation, we have:

$$\begin{array}{rcl} \ln(WAGE) & = & -0.522 \quad +0.1075EDUC \quad +0.0416EXPER \\ (se) & & (0.1986) \quad (0.0141) \quad (0.0132) \\ & & -0.0008EXPER^2 \\ & & (0.0004) \end{array}$$

- We estimate that for an additional year of education we expect wages to increase by approximately 10.75%, holding everything else constant.
- If ability has a positive effect on wages, then this estimate is overstated, as the contribution of ability is attributed to the education variable.

R Example

```
library(AER) #for `ivreg()`
library(lmtest)
library(broom)
library(POE5Rdata)
library(car) #for `hccm()` robust standard errors
library(sandwich)
library(knitr) #for making neat tables with `kable()`
library(stargazer)

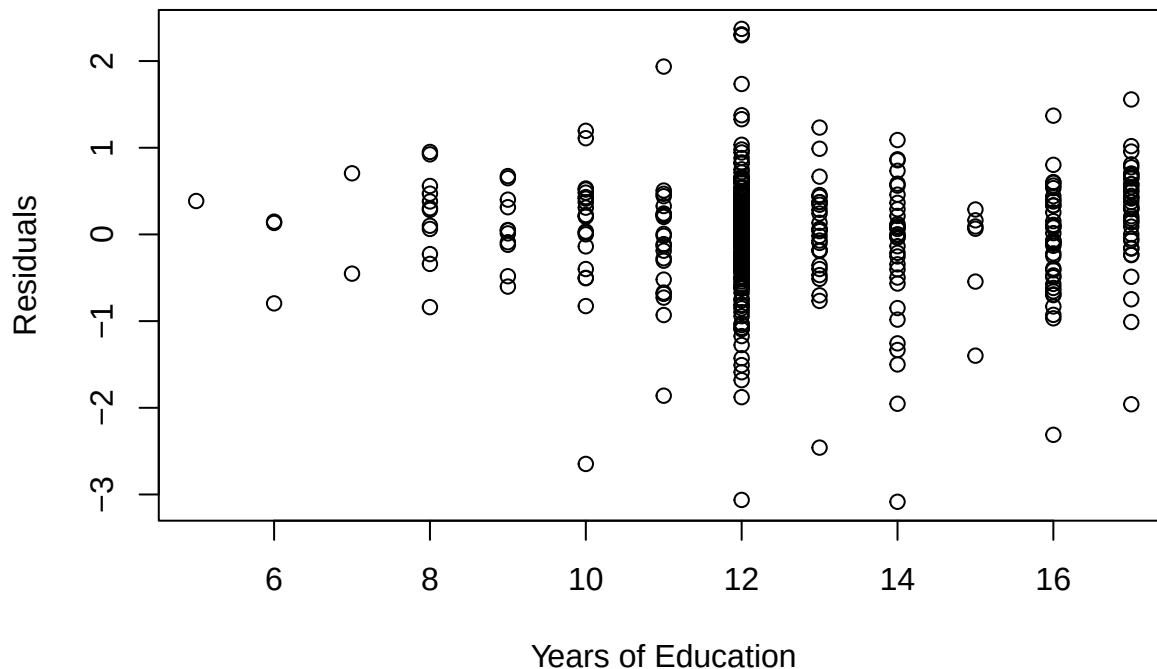
data("mroz", package="PoEdata")

mroz1 <- mroz[mroz$lfp==1,] # lfp = 1 if woman worked in 1975, else = 0
mreg.mod0 <- lm(log(wage) ~ educ + exper+I(exper^2), data=mroz1)
summary(mreg.mod0)

##
## Call:
## lm(formula = log(wage) ~ educ + exper + I(exper^2), data = mroz1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.08404 -0.30627  0.04952  0.37498  2.37115
##
## Coefficients:
```

```
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.5220406  0.1986321  -2.628  0.00890 **
## educ        0.1074896  0.0141465   7.598 1.94e-13 ***
## exper       0.0415665  0.0131752   3.155  0.00172 **
## I(exper^2)  -0.0008112  0.0003932  -2.063  0.03974 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6664 on 424 degrees of freedom
## Multiple R-squared:  0.1568, Adjusted R-squared:  0.1509
## F-statistic: 26.29 on 3 and 424 DF,  p-value: 1.302e-15
```

```
plot(mroz1$educ, mreg.mod0$residuals, ylab="Residuals", xlab="Years of Education")
```



```
# Consider the case where we omitted mother's years of education.
educ.reg <- lm(educ~exper+I(exper^2)+mothereduc, data=mroz1)
summary(educ.reg)
```

```
##
## Call:
## lm(formula = educ ~ exper + I(exper^2) + mothereduc, data = mroz1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7.4423 -1.2963 -0.0837  1.1761  5.9870
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)  9.775103   0.423889  23.061 <2e-16 ***
## exper       0.048862   0.041669   1.173  0.242
## I(exper^2)  -0.001281   0.001245  -1.029  0.304
## mothereduc  0.267691   0.031130   8.599 <2e-16 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.111 on 424 degrees of freedom
## Multiple R-squared:  0.1527, Adjusted R-squared:  0.1467
## F-statistic: 25.47 on 3 and 424 DF,  p-value: 3.617e-15
```

3 Estimators Based on the Method of Moments

- When all the usual assumptions of the linear model hold, the method of moments leads to the least squares estimator.
- If x is random and *correlated* with the error term, the method of moments leads to an alternative method, called *instrumental variables* estimation, or two-stage least squares (2SLS) estimation, that will work in large samples.
- The k th moment of a random variable Y is the expected value of the random variable raised to the k th power:

$$E(Y^k) = \mu_k = k\text{th moment of } Y \quad (7)$$

- The k th population moment in Eq. 7 can be estimated consistently using the sample (of size N) analog:

$$\hat{E}(Y^k) = \hat{\mu}_k = k\text{th sample moment of } Y = \frac{1}{N} \sum_i y_i^k \quad (8)$$

- The method of moments estimation procedure equates m population moments to m sample moments to estimate m unknown parameters.

Example:

$$\text{Var}(Y) = \sigma^2 = E[(Y - \mu)^2] = E(Y^2) - \mu^2 \quad (9)$$

- The first two population and sample moments of Y are:

Population Moments Sample Moments

$$E(Y) = \mu_1 = \mu \quad \hat{\mu} = \frac{1}{N} \sum_i y_i \quad (10)$$

$$E(Y^2) = \mu_2 \quad \hat{\mu}_2 = \frac{1}{N} \sum_i y_i^2$$

- Solve for the unknown mean and variance parameters:

$$\hat{\mu} = \frac{1}{N} \sum_i y_i = \bar{y} \quad (11)$$

and

$$\hat{\sigma}^2 = \hat{\mu}_2 - \hat{\mu}^2 = \frac{1}{N} \sum_i y_i^2 - \bar{y}^2 = \frac{1}{N} \sum_i (y_i - \bar{y})^2 \quad (12)$$

- In the linear regression model we have

$$y = \beta_1 + \beta_2 x + e,$$

- We usually assume:

$$E(e_i) = 0 \implies E(y_i - \beta_1 - \beta_2 x_i) = 0 \quad (13)$$

- If x is fixed, or random but not correlated with e , then:

$$E(x_i e_i) = 0 \implies E(y_i - \beta_1 - \beta_2 x_i) x_i = 0 \quad (14)$$

- We have two equations with two unknown parameters:

$$\begin{aligned} \frac{1}{N} \sum_i (y_i - b_1 - b_2 x_i) &= 0 \\ \frac{1}{N} \sum_i (y_i - b_1 - b_2 x_i) x_i &= 0 \end{aligned} \quad (15)$$

- These are equivalent to the least squares normal equations and their solution is:

$$\begin{aligned} b_2 &= \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2} \\ b_1 &= \bar{y} - b_2 \bar{x} \end{aligned}$$

- Under "nice" assumptions, the method of moments principle of estimation leads us to the same estimators for the simple linear regression model as the least squares principle.
- Suppose that there is another variable, z , such that:
 1. z does not have a direct effect on y , and thus, it does not belong on the right-hand side of the model as an explanatory variable.
 2. z is not correlated with the regression error term e .
 3. Variables with this property are said to be *exogenous variables*.
 4. z is strongly (or at least not weakly) correlated with x , the endogenous explanatory variable.
- A variable z with these 4 properties is called an *instrumental variable*.
- If such a variable z exists, then it can be used to form the moment condition:

$$E(ze) = 0 \implies E[z(y - \beta_1 - \beta_2 x)] = 0 \quad (16)$$

- Use Eqs. 13 and 16, the sample moment conditions are:

$$\begin{aligned} \frac{1}{N} \sum_i (y_i - \hat{\beta}_1 - \hat{\beta}_2 x_i) &= 0 \\ \frac{1}{N} \sum_i (y_i - \hat{\beta}_1 - \hat{\beta}_2 x_i) z_i &= 0 \end{aligned} \quad (17)$$

- Solving these equations leads us to the method of moments estimators, which are also called, in this case, the *instrumental variable (IV) estimators*:

$$\begin{aligned} \hat{\beta}_2 &= \frac{N \sum_i z_i y_i - \sum_i z_i \sum_i y_i}{N \sum_i z_i x_i - \sum_i z_i \sum_i x_i} \\ &= \frac{\sum_i (z_i - \bar{z})(y_i - \bar{y})}{\sum_i (z_i - \bar{z})(x_i - \bar{x})} \\ \hat{\beta}_1 &= \bar{y} - \hat{\beta}_2 \bar{x} \end{aligned} \quad (18)$$

3.1 Properties of the New Estimators:

- They are consistent, if z is exogenous, with $E(ze) = 0$.
- In large samples the instrumental variable estimators have approximate normal distributions.
- In the simple regression model:

$$\hat{\beta}_2 \sim N \left(\beta_2, \frac{\sigma^2}{r_{zx}^2 \sum_i (x_i - \bar{x})^2} \right) \quad (19)$$

- The error variance is estimated using the estimator:

$$\hat{\sigma}_{IV}^2 = \frac{1}{N-2} \sum_i (y_i - \hat{\beta}_1 - \hat{\beta}_2 x_i)^2$$

- Note that we can write the variance of the instrumental variables estimator of β_2 as:

$$Var(\hat{\beta}_2) = \frac{\sigma^2}{r_{zx}^2 \sum_i (x_i - \bar{x})^2} = \frac{Var(b_2)}{r_{zx}^2}$$

- Because $r_{zx}^2 < 1$ the variance of the instrumental variables estimator will always be larger than the variance of the least squares estimator, and thus it is said to be less efficient.
- To extend our analysis to a more general setting, consider the multiple regression model:

$$y = \beta_1 + \beta_2 x_2 + \dots + \beta_K x_K + e$$

- Let x_K be an endogenous variable correlated with the error term.
- The first $K - 1$ variables are exogenous variables that are uncorrelated with the error term e - they are called the “included” instruments.
- We can estimate this equation in two steps with a least squares estimation in each step.
- The *first stage regression* has the endogenous variable x_K on the left-hand side, and *all exogenous and instrumental variables* on the right-hand side.
- The first stage regression is:

$$x_K = \gamma_1 + \gamma_2 x_2 + \dots + \gamma_{K-1} x_{K-1} + \theta_1 z_1 + \dots + \theta_L z_L + v_K \quad (20)$$

- The least squares fitted value is:

$$\hat{x}_K = \hat{\gamma}_1 + \hat{\gamma}_2 x_2 + \dots + \hat{\gamma}_{K-1} x_{K-1} + \hat{\theta}_1 z_1 + \dots + \hat{\theta}_L z_L \quad (21)$$

- The *second stage regression* is based on the original specification:

$$y = \beta_1 + \beta_2 x_2 + \dots + \beta_K \hat{x}_K + e^* \quad (22)$$

- The least squares estimators from this equation are the *instrumental variables (IV)* estimators.
- Because they can be obtained by two least squares regressions, they are also popularly known as the *two-stage least squares (2SLS)* estimators.
- The IV/2SLS estimator of the error variance is based on the residuals from the original model:

$$\hat{\sigma}_{IV}^2 = \frac{\sum_i (y - \hat{\beta}_1 + \hat{\beta}_2 x_2 - \dots - \hat{\beta}_K x_K)^2}{N - K} \quad (23)$$

- In the simple regression, if x is endogenous and we have L instruments:

$$\hat{x} = \hat{\gamma}_1 + \hat{\theta}_1 z_1 + \cdots + \hat{\theta}_L z_L$$

- The two sample moment conditions are:

$$\begin{aligned} \frac{1}{N} \sum_i (y_i - \hat{\beta}_1 - \hat{\beta}_2 x_i) &= 0 \\ \frac{1}{N} \sum_i (y_i - \hat{\beta}_1 - \hat{\beta}_2 x_i) \hat{x}_i &= 0 \end{aligned}$$

- Solving using the fact that $\bar{\hat{x}} = \bar{x}$, we get:

$$\begin{aligned} \hat{\beta}_2 &= \frac{\sum_i (\hat{x}_i - \bar{\hat{x}}) (y_i - \bar{y})}{\sum_i (\hat{x}_i - \bar{\hat{x}}) (x_i - \bar{x})} \\ \hat{\beta}_1 &= \bar{y} - \hat{\beta}_2 \bar{x} \end{aligned}$$

- Sometimes we have more instrumental variables at our disposal than are necessary.
- Suppose we have $L = 2$ instruments, z_1 and z_2 .
- Then we have:

$$E(z_2 e) = E[z_2 (y - \beta_1 - \beta_2 x)] = 0$$

- We have three sample moment conditions:

$$\begin{aligned} \frac{1}{N} \sum_i (y_i - \hat{\beta}_1 - \hat{\beta}_2 x_i) &= \hat{m}_1 = 0 \\ \frac{1}{N} \sum_i (y_i - \hat{\beta}_1 - \hat{\beta}_2 x_i) z_{i1} &= \hat{m}_2 = 0 \\ \frac{1}{N} \sum_i (y_i - \hat{\beta}_1 - \hat{\beta}_2 x_i) z_{i2} &= \hat{m}_3 = 0 \end{aligned}$$

- The first stage regression is a key tool in assessing whether an instrument is “strong” or “weak” in the multiple regression setting.
- Suppose the first stage regression equation is:

$$x_K = \gamma_1 + \gamma_2 x_2 + \cdots + \gamma_{K-1} x_{K-1} + \theta_1 z_1 + v_K \quad (24)$$

- The key to assessing the strength of the instrumental variable z_1 is the strength of its relationship to x_K after controlling for the effects of all the other exogenous variables.
- Suppose the first stage regression equation is:

$$x_K = \gamma_1 + \gamma_2 x_2 + \cdots + \gamma_{K-1} x_{K-1} + \theta_1 z_1 + \cdots + \theta_L z_L + v_K \quad (25)$$

- We require that *at least* one of the instruments be strong.

Example: Wage Equation with IV = mother’s years of education.

- Consider the model with an instrumental variable *MOTHEREDUC*:

$$\begin{aligned} \widehat{EDUC} &= 9.7751 + 0.0489 \text{EXPER} - 0.0013 \text{EXPER}^2 \\ (se) & (0.4249) \quad (0.0427) \quad (0.0012) \\ & + 0.2677 \text{MOTHEREDUC} \\ & (0.0311) \end{aligned}$$

- To implement instrumental variables estimation using the two-stage least squares approach, we obtain the predicted values of education from the first stage equation and insert it into the log-linear wage equation to replace $EDUC$.
- Then estimate the resulting equation by least squares.
- The instrumental variables estimates of the log-linear wage equation are:

$$\ln(\widehat{WAGE}) = \begin{array}{ccc} 0.1982 & +0.0493\widehat{EDUC} & +0.0449EXPER \\ (se) & (0.0374) & (0.0136) \\ & -0.0009EXPER^2 & \\ & (0.0004) & \end{array}$$

- Note: As we can see, the coefficient of $EDUC$ is now 4.93% compared to our OLS estimate of 10.75%.
- Using $FATHEREDUC$, the first stage equation is:

$$EDUC = \gamma_1 + \gamma_2 EXPER + \gamma_3 EXPER^2 + \theta_1 MOTHEREDUC + \theta_1 FATHEREDUC + v_K$$

- The IV/2SLS estimates are:

$$\ln(\widehat{WAGE}) = \begin{array}{ccc} 0.481 & +0.0614\widehat{EDUC} & +0.0442EXPER \\ (se) & (0.0374) & (0.0136) \\ & -0.0009EXPER^2 & \\ & (0.0004) & \end{array} \quad (26)$$

- Note: In this case the coefficient of $EDUC$ is now 6.14% compared to our OLS estimate of 10.75%.
- In a multiple regression model, the coefficients are the effect of a unit change in an explanatory, independent, variable on the expected outcome, holding all other things constant.
- In calculus terminology, the coefficients are partial derivatives.
- We can net out or partial out the effects of explanatory variables.
 - Regression coefficients can be thought of measuring the effect of one variable on another after removing, or partialling out, the effects of all other variables.
 - The sample correlation between two residuals is called the *partial correlation* coefficient.
- The multiple regression model, including all m variables, is:

$$y = \underbrace{\beta_1 + \beta_2 x_2 + \cdots + \beta_G x_G}_{G \text{ exogenous variables}} + \underbrace{\beta_{G+1} x_{G+1} + \cdots + \beta_K x_K}_{B \text{ endogenous variables}} + e$$

- Think of G = Good explanatory variables, B = Bad explanatory variables and L = Lucky instrumental variables.
- It is a necessary condition for IV estimation that $L \geq B$.
- If $L = B$ then there are just enough instrumental variables to carry out IV estimation.
The model parameters are said to *just identified* or *exactly identified* in this case.
The term identified is used to indicate that the model parameters can be consistently estimated.
- If $L > B$ then we have more instruments than are necessary for IV estimation, and the model is said to be *overidentified*.

- Consider the B first-stage equations:

$$x_{G+j} = \gamma_{1j} + \gamma_{2j}x_2 + \cdots + \gamma_{Gj}x_G + \theta_{1j}z_1 + \cdots + \theta_{Lj}z_L + v_j, \quad (27)$$

for $j = 1, \dots, B$

- The predicted values are:

$$\hat{x}_{G+j} = \hat{\gamma}_{1j} + \hat{\gamma}_{2j}x_2 + \cdots + \hat{\gamma}_{Gj}x_G + \hat{\theta}_{1j}z_1 + \cdots + \hat{\theta}_{Lj}z_L$$

- In the second stage of estimation we apply least squares to:

$$y = \beta_1 + \beta_2x_2 + \cdots + \beta_Gx_G + \beta_{G+1}\hat{x}_{G+1} + \cdots + \beta_K\hat{x}_{G+B} + e^* \quad (28)$$

- Consider the model with $B = 2$:

$$y = \beta_1 + \beta_2x_2 + \cdots + \beta_Gx_G + \beta_{G+1}x_{G+1} + \beta_{G+2}x_{G+2} + e \quad (29)$$

- The first-stage equations are:

$$\begin{aligned} x_{G+1} &= \gamma_{11} + \gamma_{21}x_2 + \cdots + \gamma_{G1}x_G + \theta_{11}z_1 + \theta_{21}z_2 + v_1, \\ x_{G+2} &= \gamma_{12} + \gamma_{22}x_2 + \cdots + \gamma_{G2}x_G + \theta_{12}z_1 + \theta_{22}z_2 + v_2 \end{aligned} \quad (30)$$

- When testing the null hypothesis $H_0: \beta_k = c$, use of the test statistic

$$t = \frac{(\hat{\beta}_k - c)}{se(\hat{\beta}_k)}$$

is valid in large samples.

- It is common, but not universal, practice to use critical values, and p -values, based on the t -distribution rather than the more strictly appropriate $N(0, 1)$ distribution.
- The reason is that tests based on the t -distribution tend to work better in samples of data that are not large.
- When testing a joint hypothesis, such as $H_0: \beta_2 = c_2, \beta_3 = c_3$, the test may be based on the chi-square distribution with the number of degrees of freedom equal to the number of hypotheses (J) being tested.
 - The test itself may be called a “Wald” test, or a likelihood ratio (LR) test, or a Lagrange multiplier (LM) test.
 - These testing procedures are all asymptotically equivalent.

\end{itemize}

3.2 A word on R-squared

- Unfortunately R^2 can be negative when based on IV estimates.
- Therefore the use of measures like R^2 outside the context of the least squares estimation should be avoided.

R Example

```
library(AER) #for `ivreg()`
library(lmtest)
library(broom)
library(POE5Rdata)
library(car) #for `hccm()` robust standard errors
library(sandwich)
library(knitr) #for making neat tables with `kable()`
library(stargazer)

data("mroz", package="PoEdata")

mroz1 <- mroz[mroz$lfp==1,] # lfp = 1 if woman worked in 1975, else = 0
mreg.mod0 <- lm(log(wage) ~ educ + exper+I(exper^2), data=mroz1)
summary(mreg.mod0)
```

```
##
## Call:
## lm(formula = log(wage) ~ educ + exper + I(exper^2), data = mroz1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.08404 -0.30627  0.04952  0.37498  2.37115
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.5220406  0.1986321  -2.628  0.00890 **
## educ         0.1074896  0.0141465   7.598 1.94e-13 ***
## exper        0.0415665  0.0131752   3.155  0.00172 **
## I(exper^2)   -0.0008112  0.0003932  -2.063  0.03974 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6664 on 424 degrees of freedom
## Multiple R-squared:  0.1568, Adjusted R-squared:  0.1509
## F-statistic: 26.29 on 3 and 424 DF,  p-value: 1.302e-15
```

```
# 2SLS using 'ivreg'
# One instrument = Mother's years of education
mreg.iv1 <- ivreg(log(wage)~educ+exper+I(exper^2) |
                  exper+I(exper^2)+mothereduc, data=mroz1)
summary(mreg.iv1)
```

```
##
## Call:
## ivreg(formula = log(wage) ~ educ + exper + I(exper^2) | exper +
##       I(exper^2) + mothereduc, data = mroz1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.10804 -0.32633  0.06024  0.36772  2.34351
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept) 0.1981861 0.4728772 0.419 0.67535
## educ 0.0492630 0.0374360 1.316 0.18891
## exper 0.0448558 0.0135768 3.304 0.00103 **
## I(exper^2) -0.0009221 0.0004064 -2.269 0.02377 *
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6796 on 424 degrees of freedom
## Multiple R-Squared: 0.1231, Adjusted R-squared: 0.1169
## Wald test: 7.348 on 3 and 424 DF, p-value: 8.228e-05
```

```
# One instrument = Father's years of education
mreg.iv2 <- ivreg(log(wage)~educ+exper+I(exper^2)|
  exper+I(exper^2)+fathereduc, data=mroz1)
summary(mreg.iv2)
```

```
##
## Call:
## ivreg(formula = log(wage) ~ educ + exper + I(exper^2) | exper +
## I(exper^2) + fathereduc, data = mroz1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.09170 -0.32776  0.05006  0.37365  2.35346
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.0611170  0.4364461  -0.140  0.88870
## educ         0.0702263  0.0344427   2.039  0.04208 *
## exper        0.0436716  0.0134001   3.259  0.00121 **
## I(exper^2)   -0.0008822  0.0004009  -2.200  0.02832 *
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6719 on 424 degrees of freedom
## Multiple R-Squared: 0.143, Adjusted R-squared: 0.137
## Wald test: 8.314 on 3 and 424 DF, p-value: 2.201e-05
```

```
# Two instruments = Father's and Mother's years of education
mreg.iv3 <- ivreg(log(wage)~educ+exper+I(exper^2)|
  exper+I(exper^2)+mothereduc+fathereduc,
  data=mroz1)
summary(mreg.iv3)
```

```
##
## Call:
## ivreg(formula = log(wage) ~ educ + exper + I(exper^2) | exper +
## I(exper^2) + mothereduc + fathereduc, data = mroz1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.0986 -0.3196  0.0551  0.3689  2.3493
##
## Coefficients:
```

```
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.0481003  0.4003281   0.120  0.90442
## educ        0.0613966  0.0314367   1.953  0.05147 .
## exper       0.0441704  0.0134325   3.288  0.00109 **
## I(exper^2)  -0.0008990  0.0004017  -2.238  0.02574 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6747 on 424 degrees of freedom
## Multiple R-Squared:  0.1357, Adjusted R-squared:  0.1296
## Wald test: 8.141 on 3 and 424 DF, p-value: 2.787e-05
```

4 Specification Tests

1. Can we test for whether x is correlated with the error term? If we can it might give us a guide of when to use least squares and when to use IV estimators.
2. Can we test if our instrument is valid, and uncorrelated with the regression error, as required?
 - The null hypothesis is $H_0: Cov(x, e) = 0$ against the alternative $H_1: Cov(x, e) \neq 0$
 - If null hypothesis is true, then both the least squares estimator and the instrumental variables estimator are consistent.
 - If the null hypothesis is false, the least squares estimator is not consistent, and the instrumental variables estimator is consistent.
 - Comments:
 - Naturally if the null hypothesis is true, then one should use the more efficient estimator, which is the least squares estimator.
 - If the null hypothesis is not true, then the only consistent estimator is the instrumental variables estimator.
 - There are several forms of the test, usually called the *Hausman test*.

4.1 Hausman test–Step 1

- Consider the model:

$$y = \beta_1 + \beta_2 x + e$$

- Let z_1 and z_2 be instrumental variables for x .
- Estimate the regression model

$$x = \gamma_1 + \theta_1 z_1 + \theta_2 z_2 + v$$

by least squares, and obtain the residuals

$$\hat{v} = x - \hat{\gamma}_1 - \hat{\theta}_1 z_1 - \hat{\theta}_2 z_2 +$$

- If there are more than one explanatory variables that are being tested for endogeneity, repeat this estimation for each one, using all available instrumental variables in each regression.

4.2 Hausman test–Step 2

- Include the residuals computed in step 1 as an explanatory variable in the original regression,

$$y = \beta_1 + \beta_2 x + \delta \hat{v} + e$$

- Estimate this “artificial regression” by least squares, and employ the usual t -test for the hypothesis of significance

$$H_0 : \delta = 0 \quad \text{i.e., no correlation between } x \text{ and } e$$

$$H_1 : \delta \neq 0 \quad \text{i.e., correlation between } x \text{ and } e$$

4.3 Hausman test–Step 3

- Note that if more than one variable is being tested for endogeneity, the test will be an F -test of joint significance of the coefficients on the included residuals.

4.4 A test of the validity of the surplus moment conditions:

- **Step 1:** Compute the IV estimates $\hat{\beta}_k$ using all available instruments, including the G variables, $x_1 = 1, x_2, \dots, x_G$, that are presumed to be exogenous, and the L instruments.

- **Step 2:** Obtain the residuals

$$\hat{e} = y - \hat{\beta}_1 - \hat{\beta}_2 x_2 - \dots - \hat{\beta}_K x_K$$

- **Step 3:** Regress $\hat{e}$ on all the available instruments described in step 1.
- **Step 4:** Compute $N \times R^2$ from this regression, where N is the sample size and R^2 is the usual goodness-of-fit measure.
- **Step 5:** If all of the surplus moment conditions are valid, then

$$N \times R^2 \sim \chi^2_{(L-B)}$$

If the value of the test statistic exceeds the $100(1 - \alpha)$ -percentile from the $\chi^2_{(L-B)}$ distribution, then we conclude that at least one of the surplus moment conditions restrictions is not valid.

```
library(AER) #for `ivreg()`
library(lmtest)
library(broom)
library(POE5Rdata)
library(car) #for `hccm()` robust standard errors
library(sandwich)

# Hausman Test
data("mroz", package="POE5Rdata")
mroz1 <- mroz[mroz$lfp==1,]
attach(mroz1)

# 2SLS using 'ivreg'
# Two instruments = Mother's and Father's years of education
mreg.iv0 <- ivreg(log(wage)~educ + exper + I(exper^2)|
                  exper+ I(exper^2) + mothereduc + fathereduc, singular.ok = TRUE, data=mroz1)
summary(mreg.iv0, diagnostics=TRUE)
```

```
##
## Call:
## ivreg(formula = log(wage) ~ educ + exper + I(exper^2) | exper +
##       I(exper^2) + mothereduc + fathereduc, data = mroz1, singular.ok = TRUE)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.0986 -0.3196  0.0551  0.3689  2.3493
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.0481003  0.4003281   0.120  0.90442
## educ         0.0613966  0.0314367   1.953  0.05147 .
## exper        0.0441704  0.0134325   3.288  0.00109 **
## I(exper^2)   -0.0008990  0.0004017  -2.238  0.02574 *
##
## Diagnostic tests:
##              df1 df2 statistic p-value
## Weak instruments    2 423    55.400 <2e-16 ***
## Wu-Hausman          1 423     2.793  0.0954 .
## Sargan              1 NA      0.378  0.5386
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6747 on 424 degrees of freedom
## Multiple R-Squared:  0.1357, Adjusted R-squared:  0.1296
## Wald test: 8.141 on 3 and 424 DF, p-value: 2.787e-05

# Interpretation: (see the Colonescu online book)
# [1] Weak instruments test: rejects the null, meaning that at least one instrument
# is strong. Ho: corr(x, z) is low
# [2] Wu-Hausman test for endogeneity: barely rejects (using alpha = 10) the null that the
# variable of concern is uncorrelated with the error term, indicating that educ is
# marginally endogenous. Ho: OLS and IV are equally consistent
# [3] Sargan overidentifying restrictions: does not reject the null, meaning that
# the extra instruments are valid (are uncorrelated with the error term).
# Ho: corr (z's, e) = 0. If we reject Ho, at least one of the instruments is not valid
```

R Example

```
# Test the validity of the instruments

# Example: Cigarette consumption (data from 1995)
# P = is the after-tax average real price per pack of cigarettes in state
# Q = number of cigarette packs per capita sold
# Income = real per capita income

library(AER)
data("CigarettesSW")
# add rincome to the dataset
CigarettesSW$rincome <- with(CigarettesSW, income / population / cpi)

# compute real per capita prices
CigarettesSW$rprice <- with(CigarettesSW, price / cpi)
```

```

# compute the sales tax
CigarettesSW$salestax <- with(CigarettesSW, (taxes - tax) / cpi)

# add cigtax to the data set
CigarettesSW$cigtax <- with(CigarettesSW, tax/cpi)

c1995 <- subset(CigarettesSW, year == "1995")

# Estimate the model w/ IV = salestax
cig_m0 <- lm(log(packs) ~ log(rprice) + log(rincome) + salestax, data = c1995)
summary(cig_m0)

##
## Call:
## lm(formula = log(packs) ~ log(rprice) + log(rincome) + salestax,
##     data = c1995)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.58357 -0.08482 -0.00983  0.10160  0.39641
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  11.32364    1.38487   8.177 2.25e-10 ***
## log(rprice)  -1.67055    0.35541  -4.700 2.58e-05 ***
## log(rincome)  0.41973    0.24558   1.709  0.0945 .
## salestax      0.01444    0.01376   1.050  0.2996
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1871 on 44 degrees of freedom
## Multiple R-squared:  0.4466, Adjusted R-squared:  0.4089
## F-statistic: 11.84 on 3 and 44 DF,  p-value: 8.203e-06

# Check the instrument relevance for salestax
linearHypothesis(cig_m0, "salestax = 0", vcov = vcovHC, type = "HC1")

## Linear hypothesis test
##
## Hypothesis:
## salestax = 0
##
## Model 1: restricted model
## Model 2: log(packs) ~ log(rprice) + log(rincome) + salestax
##
## Note: Coefficient covariance matrix supplied.
##
##   Res.Df Df      F Pr(>F)
## 1      45
## 2      44  1 0.9602 0.3325

# Estimate the model w/ IV = cigtax
cig_m1 <- lm(log(packs) ~ log(rprice) + log(rincome) + cigtax, data = c1995)
summary(cig_m1)

```

```
##
## Call:
## lm(formula = log(packs) ~ log(rprice) + log(rincome) + cigtax,
##     data = c1995)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.57616 -0.08860  0.01029  0.12953  0.36070
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  12.810172   2.888506   4.435 6.07e-05 ***
## log(rprice)  -1.982728   0.678953  -2.920  0.0055 **
## log(rincome)  0.352339   0.235582   1.496  0.1419
## cigtax        0.007501   0.008208   0.914  0.3658
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1877 on 44 degrees of freedom
## Multiple R-squared:  0.4433, Adjusted R-squared:  0.4054
## F-statistic: 11.68 on 3 and 44 DF,  p-value: 9.315e-06
# Check the instrument relevance for cigtax
linearHypothesis(cig_m1, "cigtax = 0", vcov = vcovHC, type = "HC1")

## Linear hypothesis test
##
## Hypothesis:
## cigtax = 0
##
## Model 1: restricted model
## Model 2: log(packs) ~ log(rprice) + log(rincome) + cigtax
##
## Note: Coefficient covariance matrix supplied.
##
##   Res.Df Df       F Pr(>F)
## 1      45
## 2      44  1 0.7665 0.3861

# Estimate the model w/ IV1 = salestax, IV2 = cigtax
cig_m2 <- lm(log(packs) ~ log(rprice) + log(rincome) + salestax + cigtax, data = c1995)
summary(cig_m2)

##
## Call:
## lm(formula = log(packs) ~ log(rprice) + log(rincome) + salestax +
##     cigtax, data = c1995)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.55356 -0.10582 -0.01882  0.12655  0.37572
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  16.250518   3.583826   4.534 4.58e-05 ***
```

```

## log(rprice)  -2.842378    0.862505   -3.295   0.00197 **
## log(rincome)  0.481526    0.245807    1.959   0.05662 .
## salestax      0.023374    0.014842    1.575   0.12261
## cigtax        0.013128    0.008828    1.487   0.14430
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1846 on 43 degrees of freedom
## Multiple R-squared:  0.4737, Adjusted R-squared:  0.4247
## F-statistic: 9.675 on 4 and 43 DF,  p-value: 1.137e-05

# Check the instrument relevance for salestax and cigtax
linearHypothesis(cig_m2, c("salestax = 0", "cigtax = 0"), vcov = vcovHC, type = "HC1")

## Linear hypothesis test
##
## Hypothesis:
## salestax = 0
## cigtax = 0
##
## Model 1: restricted model
## Model 2: log(packs) ~ log(rprice) + log(rincome) + salestax + cigtax
##
## Note: Coefficient covariance matrix supplied.
##
##   Res.Df Df      F Pr(>F)
## 1      45
## 2      43  2 1.5397  0.226

cig_ivreg <- ivreg(log(packs) ~ log(rprice) + log(rincome) |
                  log(rincome) + salestax, data = c1995)

coeftest(cig_ivreg, vcov = vcovHC, type = "HC1")

##
## t test of coefficients:
##
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   9.43066    1.25939   7.4883 1.935e-09 ***
## log(rprice)  -1.14338    0.37230  -3.0711 0.003611 **
## log(rincome)  0.21452    0.31175   0.6881 0.494917
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

# Test the validity of the instruments (J -statistic) via
# the overidentifying restrictions test

cig_iv_OR <- lm(residuals(cig_ivreg) ~ log(rincome) + salestax + cigtax, data = c1995)

cig_OR_test <- linearHypothesis(cig_iv_OR,
                               c("salestax = 0", "cigtax = 0"),
                               test = "Chisq")
print(cig_OR_test)

## Linear hypothesis test

```

```
##
## Hypothesis:
## salestax = 0
## cigtax = 0
##
## Model 1: restricted model
## Model 2: residuals(cig_ivreg) ~ log(rincome) + salestax + cigtax
##
##   Res.Df    RSS Df Sum of Sq  Chisq Pr(>Chisq)
## 1      46 1.6172
## 2      44 1.5971  2  0.020158 0.5554    0.7575
```